

1.

(a)

(b)

2.

The Poisson distribution for a discrete $k, k = 0, 1, 2, \dots$ and real parameter λ is

$$P(k|\lambda) = e^{-\lambda} \frac{\lambda^k}{k!}.$$

(a)

Find the mean of k .

By definition, we have that

$$\begin{aligned} E[k] &= \sum_{k=0}^{\infty} k \cdot P(k|\lambda) \\ &= \sum_{k=0}^{\infty} k \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= 0 + \sum_{k=1}^{\infty} \lambda^k e^{-\lambda} \frac{k}{k!} \\ &= e^{-\lambda} \sum_{k=1}^{\infty} \lambda^k \frac{1}{(k-1)!} \end{aligned}$$

Simplifying by substituting $j = k - 1$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \lambda^{j+1} \frac{1}{j!} \\ &= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ &= \lambda e^{-\lambda} \cdot e^{\lambda} \end{aligned}$$

$$\boxed{E[k] = \lambda}$$

(b)

Find the variance of k .

We need to essentially find $E[k^2]$ here. We can quickly ascertain that directly substituting k^2 will make this hard to simplify and cancel out the k terms in both the numerator and factorial denominator, so we can instead solve for $E[k(k-1)]$:

$$\begin{aligned} E[k(k-1)] &= \sum_{k=0}^{\infty} k(k-1) \cdot e^{-\lambda} \frac{\lambda^k}{k!} \\ &= \sum_{k=0}^{\infty} e^{-\lambda} \frac{(k-1) \cdot \lambda^k}{(k-1)!} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{(k-2)!} \\ &= 0 + 0 + e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} \end{aligned}$$

Simplifying by substituting $j = k - 2$:

$$\begin{aligned} &= e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^{j+2}}{j!} \\ &= \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} \\ E[k(k-1)] &= \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2 \\ E[k^2 - k] &= \lambda^2 \\ E[k^2] - E[k] &= \lambda^2 \\ E[k^2] &= \lambda^2 + E[k] = \lambda^2 + \lambda \end{aligned}$$

We can now solve for variance:

$$\begin{aligned} \text{Var}(k) &= E[k^2] - E[k]^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ \boxed{\text{Var}(k) = \lambda} \end{aligned}$$

(c)

Find the mode of k .

The mode is the value of k with the largest probability of occurring. Since k is an integer value, it must be an integer value. We can determine this by observing the nature in which the sequence continues element by element:

$$P(k|\lambda) =$$
$$P(k+1|\lambda) = e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}$$

We see that

$$\frac{P(k|\lambda)}{P(k+1|\lambda)} = \frac{e^{-\lambda} \frac{\lambda^k}{k!}}{e^{-\lambda} \frac{\lambda^{k+1}}{(k+1)!}}$$
$$= \frac{\lambda^k \cdot (k+1)!}{\lambda^{k+1} \cdot k!}$$
$$= \frac{k+1}{\lambda}$$

So essentially, with the way this is set up, we want to find when the current probability is bigger than the next probability, since we know the Poisson distribution is parabolically shaped a maximum point after which the probability begins decreasing. We have a few different cases:

- If $\frac{k+1}{\lambda} > 1$, then probabilities are going down.
- If $\frac{k+1}{\lambda} < 1$, the probabilities are going up.
- If $\frac{k+1}{\lambda} = 1$, then $k = \lambda - 1$.

From looking at this, we can see that the highest probability or mode happens around $k = \lambda - 1$. However, since it is possible for $k = \lambda - 1$, in this case in which λ is an integer, we need to differentiate two cases, since both $\lambda - 1$ and λ have equal probability and are both integers. In the case in which λ is not an integer, then since k must be an integer, we just take λ floored to the nearest integer.

(d)

Assume two categories C_1 and C_2 , equally probably *a priori*, distributed with Poisson distributions and $\lambda_1 > \lambda_2$. What is the Bayes classification decision?

We have $P(C_1) = P(C_2) = 0.5$, and $P(x|C_1) = e^{-\lambda_1} \frac{\lambda_1^x}{x!}$ and $P(x|C_2) = e^{-\lambda_2} \frac{\lambda_2^x}{x!}$. By definition, we decide the category based on the probability comparison, since the priors are equal, i.e. we would choose C_1 if $P(C_1) > P(C_2)$, but as they are equal, we see if $P(x|C_1) > P(x|C_2)$:

$$\begin{aligned} e^{-\lambda_1} \frac{\lambda_1^x}{x!} &> e^{-\lambda_2} \frac{\lambda_2^x}{x!} \\ e^{-\lambda_1} \lambda_1^x &> e^{-\lambda_2} \lambda_2^x \\ -\lambda_1 + x \ln(\lambda_1) &> -\lambda_2 + x \ln(\lambda_2) \\ x \ln(\lambda_1) - x \ln(\lambda_2) &> -\lambda_2 + \lambda_1 \\ x(\ln(\lambda_1) - \ln(\lambda_2)) &> \lambda_1 - \lambda_2 \\ x &> \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)} \end{aligned}$$

Therefore, the decision rule is as follows:

Choose C_1 if $x > \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$; Otherwise, choose C_2 .

(e)

What is the Bayes error rate?

Logically, the Bayes error rate is the probability of making an incorrect classification; that is, either we choose C_2 when C_1 was the right category, or we choose C_1 when C_2 was the right category. Using our priors with $P(C_1) = P(C_2) = 0.5$, and letting the threshold of classification to be $y = \frac{\lambda_1 - \lambda_2}{\ln(\lambda_1) - \ln(\lambda_2)}$, with the understanding that we classify into the categories based on being above or below this threshold, we can express $P(\text{error})$ to be

$$P(\text{error}) = P(C_1) \cdot \sum_{k=0}^{\lfloor y \rfloor} P(k|C_1) + P(C_2) \cdot \sum_{k=\lceil y \rceil}^{\infty} P(k|C_2)$$

$$P(\text{error}) = \frac{1}{2} \cdot \sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \frac{1}{2} \cdot \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}$$

$$P(\text{error}) = \frac{\sum_{k=0}^{\lfloor y \rfloor} e^{-\lambda_1} \frac{\lambda_1^k}{k!} + \sum_{k=\lceil y \rceil}^{\infty} e^{-\lambda_2} \frac{\lambda_2^k}{k!}}{2}$$

3.

(a)

(b)

4.